Machine Learning

1.1 A self-driving car is an output. It uses sensors to scan the environment to avoid collisions. The censored data is input to the system, and it interprets that data using acquired knowledge already stored by the machine learning to make predictions on how to navigate the environment.

1.2 Netflix recommended system is an output. It looks at the past movies you have watched and recommends similar movies you might be interested in.

1.3 Signature recognition is also an output. It compares the signature with a signature already stored in memory. If the signature is the same the machine learning algorithm recognizes it and accepts signature. If the signature is not the same the algorithm declines.

1.4 Medical diagnosis is an output. The machine learning algorithm compares the sickness with other similar sicknesses and looks at what diagnosis were used and recommends a similar diagnosis to cure the sickness.

2. 1 Classification, because the machine learning categorizes data using predetermined labels.

2.2 Regression

2.3 Classification

2.4 Classification

3.1 Unsupervised learning the machine learning algorithm has no historical knowledge it leans and predicts as it goes

3.2 Supervised learning the machine learning algorithm looks at historical transactions and customer demographics to predict customer lifetime value.

3.3 Supervised learning because it uses purchase history, browsing behavior and preferences as input features to predict the output of segmenting customer demographics.

3.4 Unsupervised the algorithm has no historical social media posts to refer to.

4.1 Not appropriate to use semi-supervised, as semi-supervised learning uses a small portion of labelled data to guide the learning process. In this case it uses a large portion of labelled data to guide the learning process.

4.2 Appropriate to use semi-supervised as a small portion of labelled data is used to guide the learning process.

4.3 Not appropriate to use semi-supervised as most of the emails are labelled, whereas semi-supervised is suitable for a minor set of labelled data to guide the learning process.

4.4 Not appropriate as complete financial and credit-related information is labelled data